MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

WEEKLY CHAPTER TEST – 05
Chapter 13: Surface Areas and Volumes

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 objective questions (1 mark each).
2. Section B has 2 Short Answer questions (2 marks each). Section C has 2 questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (6 marks). Show all steps clearly.
4. Take $\pi = \frac{22}{7}$ unless specified otherwise. Show all steps and calculations clearly.

SECTION A: MULTIPLE CHOICE QUESTIONS (1 Mark Each) [4 Marks]

1. A cylindrical bucket of height 32 cm and base radius 18 cm is filled with sand. When emptied on the ground, a conical heap of height 24 cm is formed. The radius of the base of the heap is: **[1]**

(a) (a) 12 cm (b) 24 cm (c) 36 cm (d) 48 cm

2. The curved surface area of a right circular cone of height 15 cm and base diameter 16 cm is: **[1]**

(a) (a) 60π cm^2 (b) 68π cm^2 (c) 120π cm^2 (d) 136π cm^2

3. A right triangle with perpendicular sides 3 cm and 4 cm is revolved about the side of 3 cm to form a cone. The volume of the cone so formed is: **[1]**

(a) (a) 12π cm^3 (b) 15π cm^3 (c) 16π cm^3 (d) 20π cm^3

4. **Assertion (A):** If the radius of the base of a right circular cylinder is halved and its height is doubled, its volume is halved.
Reason (R): If the radius of the base of a right circular cone of volume V is halved and its height is doubled, the volume of the new cone becomes $\frac{V}{2}$. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

5. A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends. The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area. **[2]**

6. Mayank made a bird-bath for his garden in the shape of a cylinder with a hemispherical depression at one end. The height of the cylinder is 1.45 m and its radius is 30 cm. Find the total surface area of the bird-bath. **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

7. A wooden article was made by scooping out a hemisphere from each end of a solid cylinder. If the height of the cylinder is 10 cm and its base is of radius 3.5 cm, find the total surface area of the article. **[3]**

8. A cone of maximum size is carved out from a solid wooden cube of edge 14 cm. Find the surface area of the cone and the surface area of the remaining solid left out after the cone is carved out. **[3]**

SECTION D: LONG ANSWER / HOTS QUESTION (6 Marks) [6 Marks]

9. A circus tent is in the shape of a cylinder surmounted by a conical top of the same diameter. Their common diameter is 56 m, the height of the cylindrical part is 6 m, and the total height of the tent above the ground is 27 m.

(i) Find the slant height of the conical portion.
(ii) Find the total area of the canvas fabric used in making the tent.
(iii) Find the cost of canvas bought at the rate of 200 per m^2, if 30 m^2 canvas was wasted in stitching.

[6]

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— **END OF QUESTION PAPER** —

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